

Third Terminal Examination – 2082

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

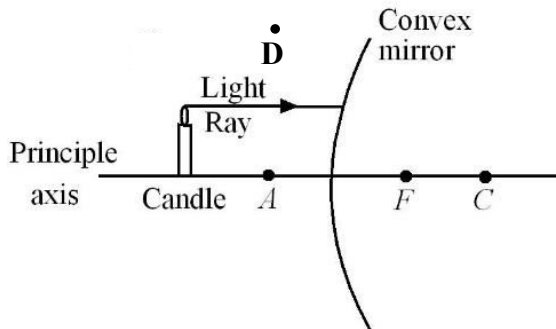
Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. $[11 \times 1 = 11]$

- Dimensional formula of Latent Heat is:
 - $[M^0 L^1 T^{-2}]$
 - $[M^0 L^2 T^{-2}]$
 - $[M^0 L^2 T^{-2} K^{-1}]$
 - $[M^0 L^2 T^{-2} K^{-2}]$
- The forces $\vec{F}_1 = (3\hat{i} + 4\hat{j})\text{N}$ and $\vec{F}_2 = (4\hat{i} + 3\hat{j})\text{N}$ are acting on a body. The resultant force on the body is:
 - 10 N
 - 14 N
 - $5\sqrt{2}$ N
 - $7\sqrt{2}$ N
- Prinsha goes to her school at a speed of 40 km/hr and returns back at 60 km/hr along the same path. Her average speed is:
 - 36 km/hr
 - 40 km/hr
 - 45 km/hr
 - 48 km/hr
- A machine gun of mass M fires n bullets per second each of mass m at velocity v . The initial acceleration of the gun with recoil speed V will be:
 - $\frac{MV}{nm}$
 - $\frac{mV}{nM}$
 - $\frac{mnv}{M}$
 - $\frac{Mnv}{m}$
- Bishnu sir is very hardworking person in our college. His mass is 60 kg and he runs up 30 steps in 40 s, in going from one floor to another. If each step is 20 cm high, calculate the power of Bishnu sir.
 - 6 W
 - 60 W
 - 88.2 W
 - 30.5 W

6. Two bodies will be in thermal equilibrium if they have same
- Specific heat
 - Heat energy
 - Temperature
 - Thermal conductivity
7. The diagram alongside shows a light ray parallel to the principal axis of a spherical convex (diverging) mirror. Point F is the virtual focal point of the mirror and C is the centre of curvature. After the light ray reflected, it will pass through point



- A
 - C
 - D
 - F
8. A, B and C are three optical media of respective critical angles C_1 , C_2 and C_3 . Total internal reflection can occur from A to B and also from B to C but not from C to A. The correct relation for the critical angles will be
- $C_1 > C_2 > C_3$
 - $C_1 = C_2 = C_3$
 - $C_3 > C_1 > C_2$
 - $C_1 < C_2 < C_3$
9. NC^{-1} has the same unit as
- Volt – meter
 - Volt/meter
 - Farad – m
 - Farad/m
10. The work done in moving charge $2C$ from one point to another point having potential difference $75V$ is
- 0 J
 - 75 J
 - 150 J
 - 300 J
11. A Parallel plate capacitor has capacitance $5\mu F$ in air and $15\mu F$ in oil. The dielectric constant of oil is
- $1/3$
 - 3
 - $3/2$
 - $2/3$

Group – B

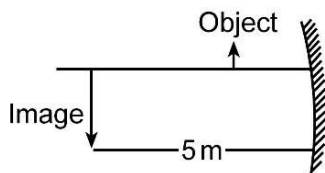
Give short answer to the following questions.

[8 × 5 = 40]

12. a) Under what condition the scalar product of two vectors is equal to $\sqrt{3}$ times the magnitude of their cross product? [1]
- b) Distance 's' travelled by a particle at any instant 't' can be represented as $s = B(t+C) + At^2$. Find the dimensions of A and C. [2]
- c) Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{C} = \vec{A} + \vec{B}$ and $C^2 = A^2 + B^2$. Find angle between $\vec{A}$ and $\vec{B}$. [2]
13. a) Using v-t graph, derive the relation $v = u + at$, where the symbols carry their usual meaning. [2]
- b) A ball is thrown vertically upward from ground with velocity of 20 m/s. Calculate:
- i) Maximum height reached by ball. [1]
- ii) Time taken by ball to return to ground. [1]
- ii) Sketch the nature of velocity – time graph for the motion of the ball. [1]
14. a) Show that net workdone on body is equal to change in its kinetic energy. [2]
- b) A water reservoir tank of 250 m^3 is situated at height of 20m from the water level. What will be the power of the electric motor to be used to fill the tank in three hours. Efficiency of motor is 70%. [3]
15. a) Why a column of mercury in a thermometer first descends slightly and then rises when placed in hot water? Explain. [1]
- b) Find the relationship among real and apparent expansivity of liquid with expansivity of vessel. [2]
- c) A copper wire of diameter 0.5 mm is stretched between two points at 25°C . Calculate the increase in tension in the wire if the temperature falls to 0°C . (Young's modulus for copper = $1.2 \times 10^{11} \text{ Nm}^{-2}$, linear expansivity for copper = $18 \times 10^{-6} \text{ K}^{-1}$) [2]

16. a) What is specific latent heat of vaporization of a liquid? [1]
 b) Develop an expression for the determination of the latent heat of vaporization of water. [2]
 c) What is the result of mixing 10gm of ice at 0°C into 15gm of water at 20°C in an iron vessel of mass 100 gm. (specific heat capacity of iron is $0.09 \text{ cal gm}^{-1} \text{ }^{\circ}\text{C}^{-1}$ and latent heat of fusion of ice is 80 cal gm^{-1}) [2]

17. a) Define lateral shift. Write the condition when the lateral shift becomes zero. [2]
 b) A certain projector uses a concave mirror for projecting an object's image on a screen. It produces an image that is 5 times bigger than the object and the screen is 5m away from the mirror as shown in Fig.

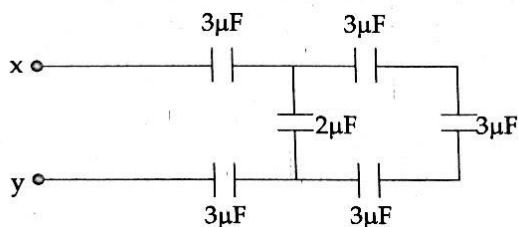


- i. Give reason why is the image larger than the object? [1]
 ii. Calculate the focal length of the mirror. [2]
18. a) What is a prism? Prove that the relation, $A + \delta = i + e$ for a prism, where the symbols have their usual meanings. [1+2]
 b) What do you mean by I-D curve in prism? [1]
 c) Sketch a graph to show variation of angle of deviation with angle of incidence. [1]
19. a) Define the capacitance of a capacitor. What are the factors on which the capacitance of parallel plate capacitor depends? [1+1]
 b) i. Two plane parallel conducting plates 15.0 mm apart are held horizontal, one above the other, in air. The upper plate is maintained at a positive potential of 1500 V while the lower plate is earthed. Calculate the number of electrons which must be attached to a small oil drop of mass $4.9 \times 10^{-15} \text{ kg}$ if it remains stationary in the air between the plate. (Assume that the density of air is negligible in comparison with that of oil). [2]

- ii. If the potential of the upper plate is suddenly changed to -1500 V .
What is the initial acceleration of the charged drop? [1]

OR

- If the electric field is zero throughout a certain region of space, is the potential also zero in the region or not? Explain. [2]
- What is the physical significance of relative permittivity of a material placed between two plates of a capacitor? [1]
- Compute the equivalent capacitance of the network between points x and y. [2]



Group – C

Give long answer to the following questions.

[3 × 8 = 24]

- For a projectile projected horizontally from 'H' height above the ground, find the expression for Time of flight and Horizontal Range. [2]
 - If you drop a ball of certain mass from a height 'h' and at the same instant another ball of same mass is thrown horizontally with a velocity 'u', which ball will hit the ground first and why? [1]
 - A body is projected horizontally from the top of a tower 100 m high with a velocity of 9.8 m/s . Find the velocity with which it hits the ground. [2]
 - Is there any example in which the direction of velocity of a body changes but its acceleration remains constant? [1]
 - A person moving in a rainy day must holds umbrella at a certain angle though the rain is falling vertically down. Why? [2]

OR

- State the laws of limiting friction. Show that tangent of angle of friction is equal to coefficient of friction. [1+2]
- Why does a cricketer lowers his hand while catching a ball? [1]

- c) What is the acceleration of the block sliding down a 30° inclined plane when the coefficient of friction is 0.25? [2]
- d) A 55 kg man is standing on the bathroom scale inside elevator. The elevator is moving upward with acceleration of 2 m/s^2 . Draw free-body diagram of problem and find the reading shown by scale. [2]

21. a) i. Define thermal conductivity. State the factors on which it depends. [1+1]

ii. Iron chair appears to be more cold than wooden chair in winter. Explain. [1]

- b) How much heat per day is lost by conduction through a concrete wall, whose area is 100 m^2 and thickness is 25 cm. The inside temperature is 25°C and outside is 0°C .

[Thermal conductivity for concrete is $0.42 \text{ Wm}^{-1}\text{K}^{-1}$] [2]

- c) The marking on an aluminum ruler and a brass ruler are perfectly aligned at 0°C . How far apart will the 20.0 cm marks be on the two rulers at 100°C , if precise alignment of the left-hand ends of the rulers is maintained? [2]

[The coefficient of linear expansion of aluminum and brass are $24 \times 10^{-6} \text{ K}^{-1}$ and $20 \times 10^{-6} \text{ K}^{-1}$ respectively.]

22. a) State the Gauss law in electrostatics. [1]

- b) Derive an expression for electric field intensity due to linear charged conductor. [3]

- c) Derive an expression for energy stored in charged capacitor. [2]

- d) What distance must an electron move in a uniform potential gradient 200 V cm^{-1} in order to gain kinetic energy $3.2 \times 10^{-18} \text{ J}$? [2]

☞ - The End - ☞